Machine Learning-Based Fuel Consumption Prediction and Cruise Fuel-Efficiency Analysis for Airbus A321

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1 Introduction and Executive Summary

1.1 Project motivation

Aircraft fuel consumption is strongly affected by the conditions under which a flight is operated. Altitude, speed, climb or descent behavior, and the distance covered during a trajectory interval can all be associated with changes in fuel burn. A data-driven model can therefore be useful for studying how operational flight characteristics relate to observed fuel consumption.

This project focuses specifically on the Airbus A321 and investigates two related questions:

1. Can machine learning predict fuel consumption rate from trajectory and operational variables?
2. Within the observed cruise region, can the fitted model be used to explore combinations of altitude and groundspeed associated with lower observed or predicted fuel-use rates?

The project uses the EUROCONTROL Performance Review Commission Aircraft Fuel Burn Estimation Data Challenge 2025 dataset. The dataset combines ADS-B trajectory information, ACARS fuel telemetry, and flight metadata. The published dataset contains approximately 15,000 flights, multiple aircraft types, and interval-level fuel information. The present analysis deliberately restricts the modeling population to A321 flights. The dataset and its description are available from Zenodo.

1.2 Data subset

The original training flight list contained 1.1037×10^4 flights. There were 285 flights identified as aircraft type A321, and all 285 of these A321 flights had corresponding fuel telemetry records.

After matching fuel intervals to trajectory observations, 2572 A321 fuel intervals were available and 2336 of them could be matched to trajectory data. A quality filter requiring at least 90% trajectory coverage was then applied. The final modeling data contained 2273 interval-level observations from 283 flights.

The main response variable is **fuel rate**, defined as fuel consumed per minute:

$$\text{Fuel rate} = \frac{\text{fuel consumed (kg)}}{\text{interval duration (min)}}.$$

The principal predictors used in the final models were:

- distance covered during the interval (`distance_km`);
- mean altitude (`mean_altitude`);
- altitude change (`altitude_change`);
- derived groundspeed;
- derived vertical rate;
- flight phase.

The three flight phases used in the final modeling data were **Climb**, **Cruise**, and **Descent**. Intervals classified as `Unknown` were retained for exploratory analysis but removed from the final prediction models.

1.3 Executive summary of findings

The final dataset contained a median fuel-use rate of 40.079 kg/min and a mean of 44.607 kg/min. The distribution was strongly affected by high values in short intervals, particularly during climb.

Three predictive algorithms were compared:

- linear regression;
- random forest;
- gradient boosting.

A flight-level rather than row-level split was used to reduce information leakage between training and validation data. The validation set contained 57 previously unseen flights.

Among the tested models, the random forest obtained a validation RMSE of **19.77 kg/min**, an MAE of **10.05 kg/min**, and an R^2 of **0.368**. The same model remained the best-performing model in the comparison after a robustness check that restricted attention to intervals of at least one minute.

The cruise analysis was then performed only in regions with substantial observed support. The out-of-sample validation analysis identified low observed fuel-rate cells around approximately 35,500–36,000 ft and 410–420 groundspeed. These values describe patterns present in this dataset; they are not presented as operational flight instructions.

2 Methods and Analysis

2.1 Data preparation and quality control

The data preparation pipeline combined three components:

1. the training flight list, containing aircraft type and flight metadata;
2. the fuel telemetry intervals, containing interval-level fuel consumption;
3. the trajectory Parquet files, containing timestamped position and flight-state variables.

The main quality-control rules were:

- keep A321 flights only;
- match fuel records to flight identifiers;
- match each fuel interval to trajectory points;
- calculate a trajectory coverage ratio;
- retain intervals with at least 90% coverage;
- calculate the fuel rate from interval fuel and duration;
- avoid using predictor variables with very high missingness, such as the raw CAS, TAS, and Mach fields;
- impute the remaining missing derived operational variables using training-set medians only.

The 90% coverage criterion retained the large majority of useful intervals while reducing the influence of poorly observed intervals.

2.2 Response variable and derived variables

Fuel consumption was recorded as fuel mass in kilograms for each telemetry interval. Because interval duration varied substantially, the primary response variable for prediction was the normalized rate in kg/min.

Derived variables were used when direct trajectory fields were incomplete. In particular:

$$\text{Derived groundspeed} = \frac{\text{distance covered}}{\text{interval duration}}$$

and

$$\text{Derived vertical rate} = \frac{\text{altitude change}}{\text{interval duration}}.$$

These derived quantities were calculated before modeling and were used as predictors because they provided complete or substantially more complete information than the raw variables.

2.3 Flight-phase classification

Flight phase was determined from mean vertical rate:

- **Climb:** mean vertical rate > 500;
- **Descent:** mean vertical rate < -500;
- **Cruise:** otherwise;
- **Unknown:** missing vertical-rate information.

This rule is a pragmatic classification designed for interval-level analysis. It should not be interpreted as a certified operational definition of flight phase.

2.4 Exploratory data analysis

Several exploratory analyses were performed before model fitting.

First, the overall distribution of fuel rate was examined. Second, fuel rate was compared across flight phases. Third, relationships between fuel rate and altitude, groundspeed, and interval duration were visualized. Finally, pairwise correlations among numeric variables were examined.

The main descriptive values were:

Table 1: Descriptive statistics for the final A321 modeling data.

Variable	Minimum	Median	Mean	Maximum
Fuel consumption (kg)	0.454	199.000	302.146	3311.222
Fuel rate (kg/min)	1.421	40.079	44.607	510.151
Interval duration (min)	0.029	4.998	7.292	59.050
Mean altitude (ft)	-106.400	34009.100	30101.700	39015.600
Mean groundspeed	124.500	429.200	409.300	536.000

A large proportion of intervals were short:

Table 2: Number of short trajectory/fuel intervals.

Duration threshold	Number of observations
< 1 minute	165

Duration threshold	Number of observations
< 5 minutes	1180
< 10 minutes	1732

There were 165 intervals shorter than one minute, 1,180 shorter than five minutes, and 1,732 shorter than ten minutes. This matters because dividing fuel by a very short duration can create unusually large fuel-rate values.

Table 3: Examples of high fuel-rate observations.

Observation	Interval_min	Fuel_kg	Fuel_rate_kg_min	Phase
Highest rate	0.96	490	510.15	Climb
Second highest	0.52	182	347.99	Climb
Third highest	0.69	236	344.31	Climb
A long interval with high rate	14.60	2620	179.40	Unknown

The highest observed rate was about 510.15 kg/min in a 0.96-minute climb interval. Several of the other largest values also came from short climb intervals. This suggests that short-duration intervals should be interpreted cautiously rather than treated as representative steady-state fuel-use conditions.

2.5 Fuel rate by flight phase

Table 4: Fuel-rate statistics by flight phase.

Phase	Observations	Mean (kg/min)	Median (kg/min)	SD	Minimum	Maximum
Climb	221	90.5	79.0	52.5	6.10	510.15
Cruise	1503	42.9	40.9	17.9	1.94	252.00
Descent	414	27.4	26.7	14.4	1.42	137.00
Unknown	135	41.7	42.9	27.3	1.90	253.00

The mean fuel rate was approximately 90.5 kg/min during climb, 42.9 kg/min during cruise, and 27.4 kg/min during descent. The phase-level differences are large, which supports including flight phase as a predictor in the regression models.

However, the standard deviations show that substantial within-phase variation remains. For example, the climb phase had a standard deviation of about 52.5 kg/min. Therefore, a simple phase mean is not sufficient to fully describe interval-level fuel consumption.

2.6 Exploratory figures

2.6.1 Fuel-rate distribution

The distribution is concentrated around moderate fuel rates but contains a long upper tail. The tail is partly associated with very short intervals, especially during climb.

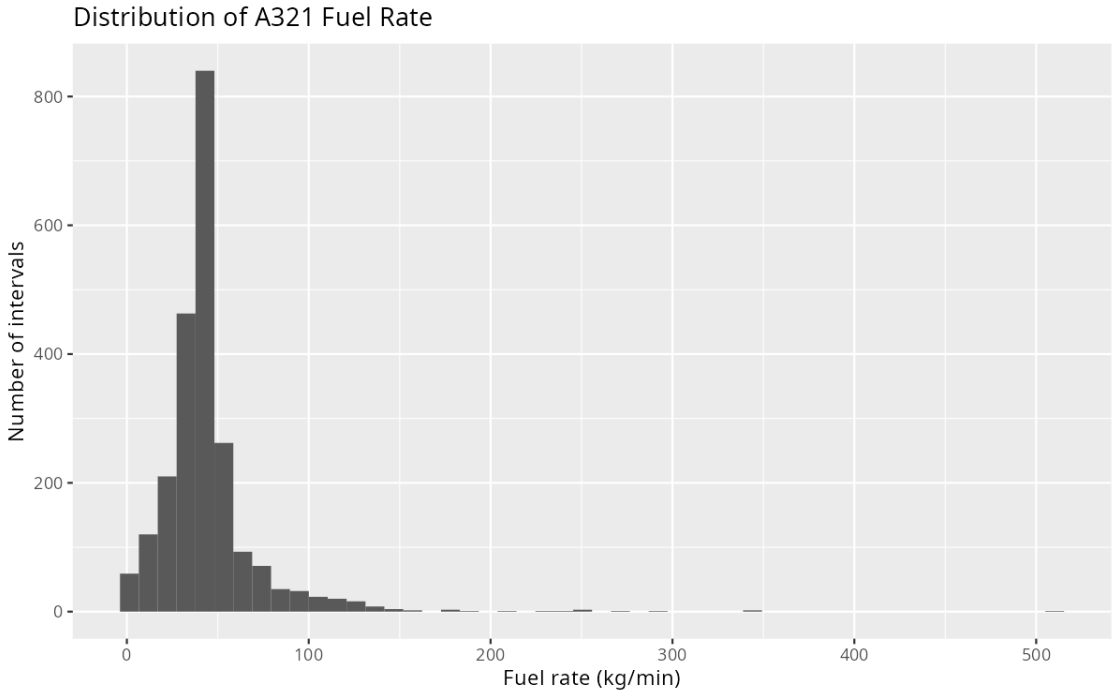


Figure 1: Distribution of fuel consumption rate in the final A321 modeling data.

2.6.2 Fuel rate by flight phase

The phase comparison reinforces the descriptive results: climb intervals generally show higher rates, while descent intervals generally show lower rates.

2.6.3 Fuel rate and altitude

The relationship between fuel rate and altitude is not well described by a simple linear trend across the full dataset. In particular, many cruise observations cluster at high altitude, while climb and descent observations occupy broader vertical ranges.

2.6.4 Fuel rate and groundspeed

Groundspeed shows considerable overlap between observations with different fuel rates. This motivates a multivariable model rather than attempting to infer fuel consumption from speed alone.

2.6.5 Fuel rate and interval duration

Very short intervals are associated with greater volatility in fuel-rate estimates. This is one reason a later robustness analysis restricted attention to intervals of at least one minute.

2.7 Correlation analysis

Several associations were observed in the exploratory analysis.

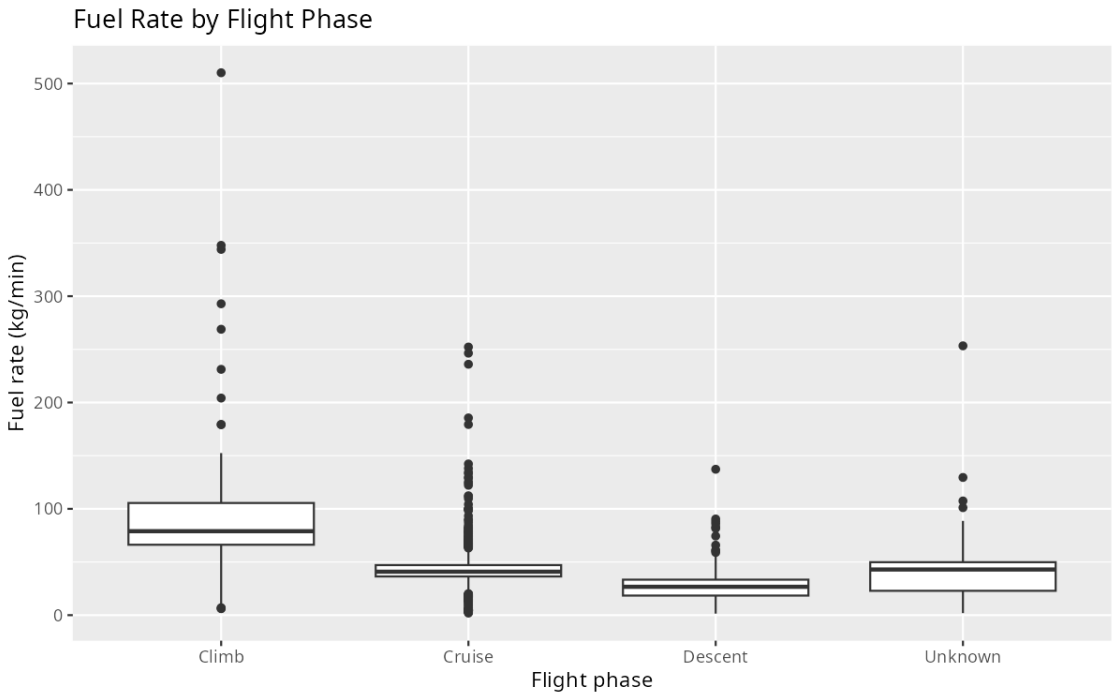


Figure 2: Fuel consumption rate grouped by flight phase.

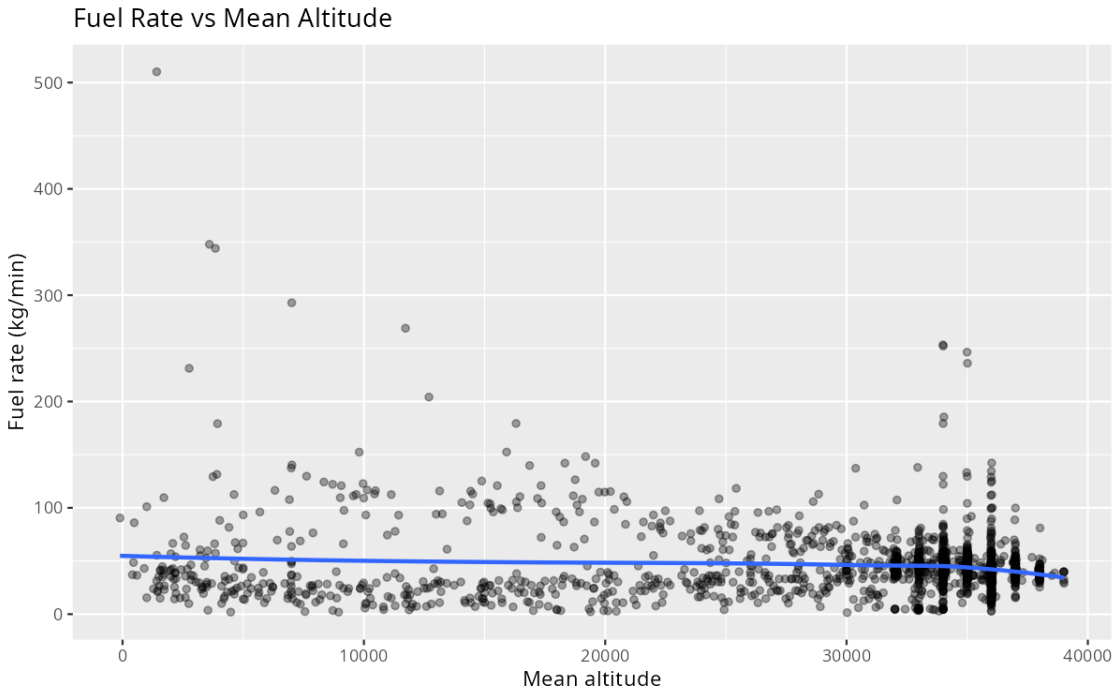


Figure 3: Fuel consumption rate as a function of mean altitude.

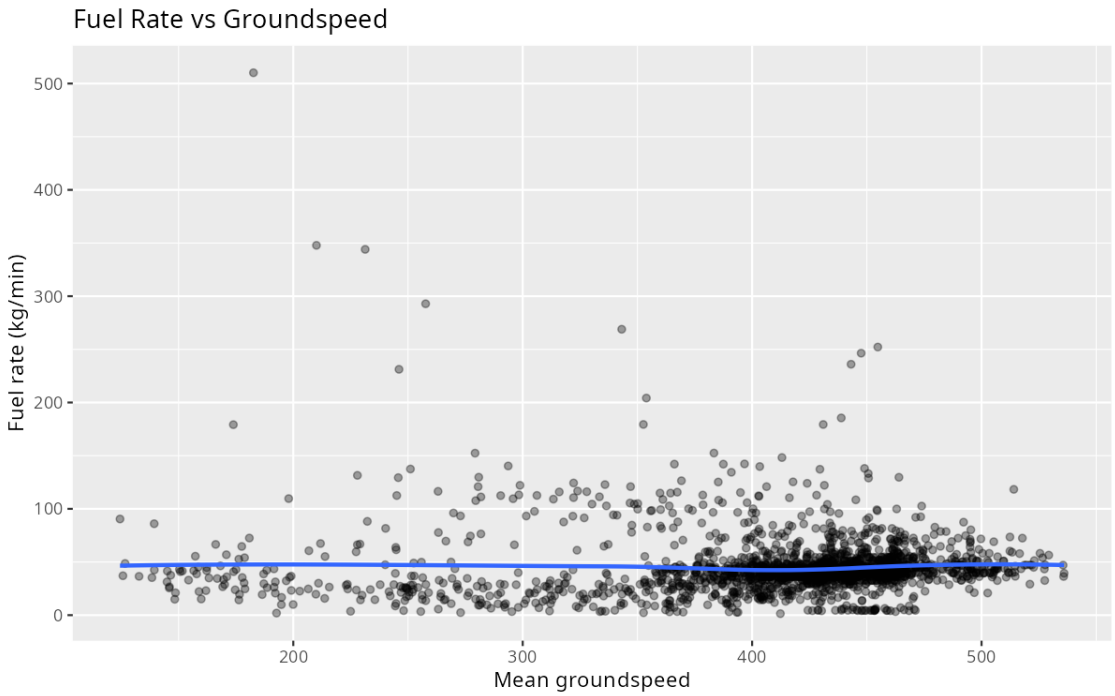


Figure 4: Fuel consumption rate as a function of mean groundspeed.

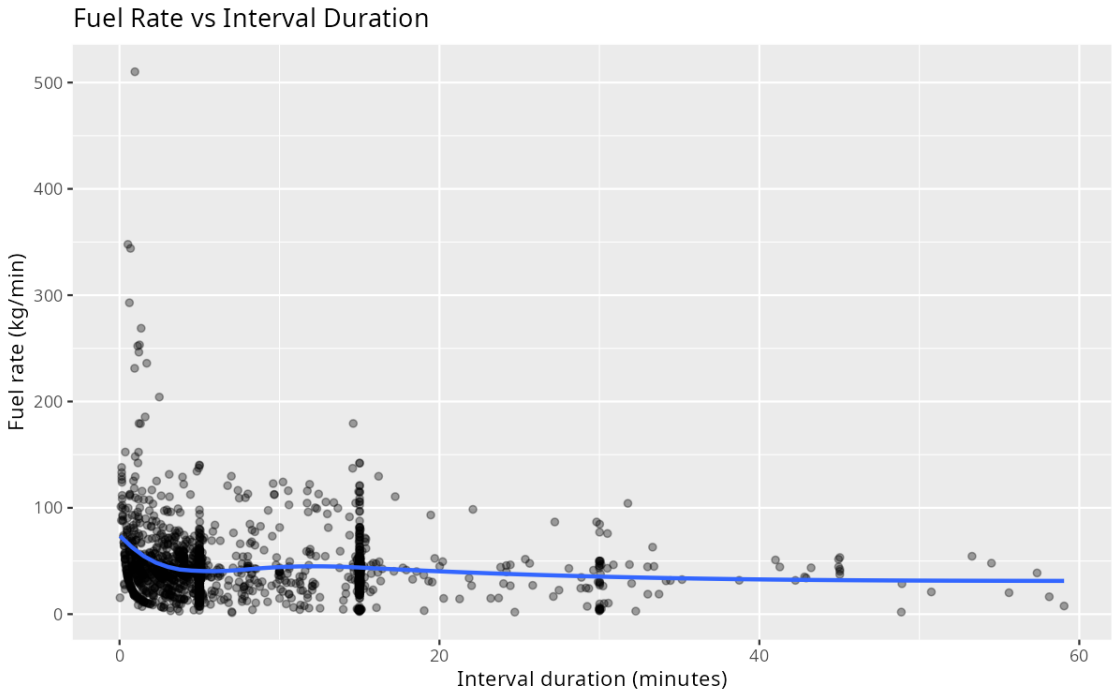


Figure 5: Fuel consumption rate as a function of interval duration.

The correlation between fuel rate and mean vertical rate was approximately **0.582**, while the correlation between fuel rate and altitude change was approximately **0.421**. By comparison, the correlation between fuel rate and distance was approximately **-0.038**, and the correlation between fuel rate and mean altitude was approximately **-0.107**.

These values indicate that short-term changes in the vertical state are more directly associated with interval fuel-rate variation than distance alone in this particular sample.

Correlation does not imply causation. In addition, some predictors such as mean altitude, minimum altitude, and maximum altitude are highly correlated with one another, so model interpretation should focus on predictive contribution rather than causal claims.

3 Predictive Modeling

3.1 Train-validation design

The final modeling data contained 283 flights.

Rather than randomly splitting individual rows, flights were divided into:

- **226 training flights**;
- **57 validation flights**.

All intervals from a given flight were kept in the same partition. This design prevents intervals from the same flight appearing in both training and validation sets and therefore provides a more realistic estimate of performance on unseen flights.

The models were fitted using the following predictors:

```
distance_km
mean_altitude
altitude_change
derived_groundspeed
derived_vertical_rate
flight_phase
```

3.2 Baselines

Two simple baselines were evaluated:

1. predicting the overall training-set mean;
2. predicting the training-set mean for each flight phase.

The overall mean baseline produced an RMSE of **24.91 kg/min** and an MAE of **15.01 kg/min**. The flight-phase baseline reduced the RMSE to **21.47 kg/min** and the MAE to **11.78 kg/min**.

These baselines establish that modeling flight phase alone explains part of the variation, but additional predictor variables are needed for better interval-level prediction.

3.3 Linear regression

The first predictive model was multiple linear regression:

$$y = \beta_0 + \beta_1 \text{distance} + \beta_2 \text{altitude} + \beta_3 \text{altitude change} + \beta_4 \text{speed} + \beta_5 \text{vertical rate} + \beta_6 \text{phase} + \epsilon.$$

The model achieved a validation RMSE of **21.05 kg/min**, MAE of **11.24 kg/min**, and $R^2 = 0.285$.

Linear regression provides a useful reference model because its coefficients are relatively easy to interpret. However, the exploratory analyses suggested nonlinear relationships and interactions, motivating the use of tree-based ensemble methods.

3.4 Random forest

The second model was a random forest with 500 trees. Random forests can represent nonlinear relationships and interactions without requiring them to be specified explicitly.

The model achieved:

- RMSE = **19.77 kg/min**;
- MAE = **10.05 kg/min**;
- $R^2 = 0.368$.

Relative permutation importance (%IncMSE) was:

Table 5: Random-forest variable importance.

Feature	Percent increase in MSE
altitude_change	25.12
distance_km	22.16
derived_vertical_rate	21.64
derived_groundspeed	17.05
mean_altitude	16.11
flight_phase	14.73

The largest importance values were associated with altitude change, distance, and derived vertical rate. This indicates that the model used both movement through the airspace and interval-level flight-state changes when making predictions.

Importance measures are predictive diagnostics rather than causal evidence.

3.5 Gradient boosting

The third model was a gradient-boosting model using the `gbm` package. The final model used a cross-validated estimate of the number of trees, with the best tree count selected from the training data.

The validation performance was:

- RMSE = **20.23 kg/min**;
- MAE = **10.48 kg/min**;
- $R^2 = 0.339$.

The strongest relative influence was associated with derived vertical rate, followed by distance and altitude change.

3.6 Model comparison

Table 6: Validation-set comparison of predictive models.

Model	RMSE (kg/min)	MAE (kg/min)	R-squared
Overall mean	24.91	15.01	NA
Flight-phase mean	21.47	11.78	NA
Linear regression	21.05	11.24	0.285
Random forest	19.77	10.05	0.368
Gradient boosting	20.23	10.48	0.339

The random forest produced the lowest validation RMSE among the models tested. Its RMSE was about 19.77 kg/min, compared with 21.05 kg/min for linear regression and 20.23 kg/min for gradient boosting.

The result indicates that nonlinear tree-based modeling captured part of the structure that a linear model did not represent. However, the R^2 value of 0.368 also shows that substantial variation in interval-level fuel rate remains unexplained.

4 Model Tuning and Diagnostics

4.1 Random-forest tuning

A grouped five-fold cross-validation procedure was used on the training flights to examine several random-forest settings.

The tested grid varied `mtry` from 2 to 5 and `nodesize` between 5 and 10. The best configuration within the grid was:

- `mtry = 2`;

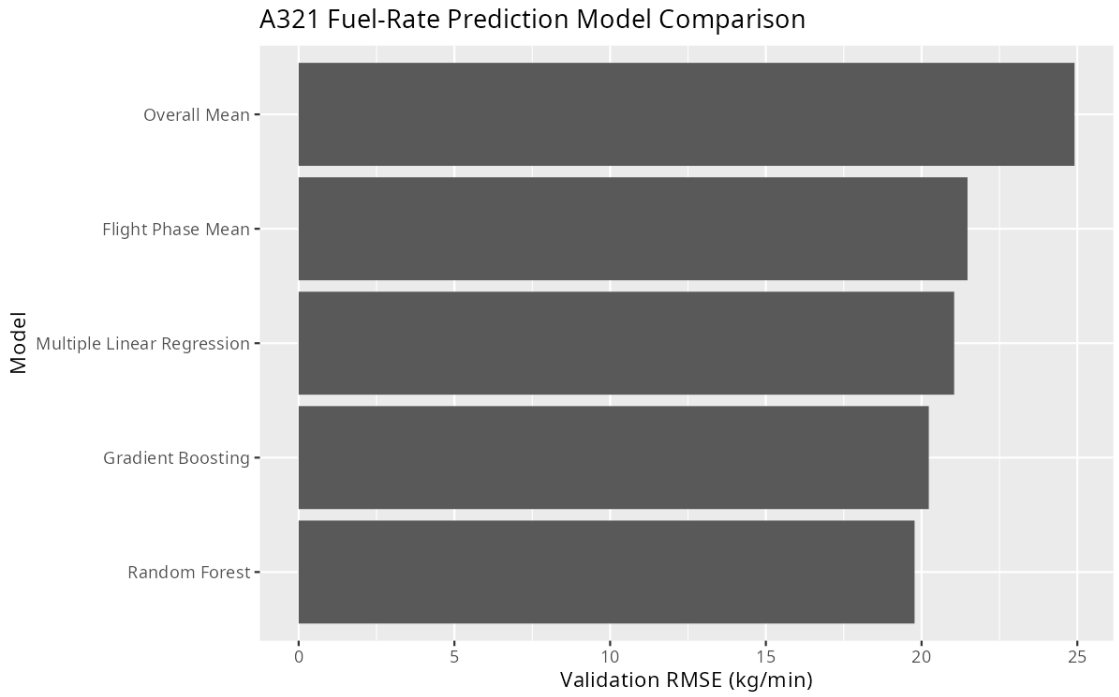


Figure 6: Validation RMSE comparison across the main predictive models and baselines.

- `nodesize = 5`.

The mean cross-validation RMSE for this configuration was approximately 20.55 kg/min. When the tuned model was retrained using all training flights and evaluated on the untouched validation flights, its validation performance remained essentially unchanged from the original random forest.

This indicates that the original random-forest configuration was already close to the best configuration in the tested parameter grid.

4.2 Robustness to short intervals

A second validation check restricted the validation intervals to duration of at least one minute. This reduced the influence of extremely short intervals.

Table 7: Model performance on validation intervals of at least one minute.

Model	RMSE (kg/min)	MAE (kg/min)	R-squared
Linear regression	19.19	9.89	0.316
Random forest	18.06	8.82	0.394
Gradient boosting	18.26	9.01	0.381

The random forest remained the lowest-RMSE model in this robustness analysis, with an RMSE of approximately **18.06 kg/min**.

This consistency suggests that the main model comparison is not driven solely by the most extreme, very short-duration observations.

4.3 Error diagnostics

The random-forest validation results showed different error behavior by flight phase:

- Climb: RMSE about 18.5 kg/min;
- Cruise: RMSE about 21.7 kg/min;
- Descent: RMSE about 11.1 kg/min.

The model also had a mean residual of about **+2.42 kg/min**, indicating slight overall underprediction on the validation set.

Error magnitude increased considerably in the high fuel-rate tail. For observations with fuel rates of at least 100 kg/min, the validation RMSE was approximately 82.9 kg/min. This highlights the difficulty of predicting rare high-rate events from the limited A321 sample.

These diagnostics suggest that the model describes the central portion of the data more reliably than the extreme tail.

5 Cruise Fuel-Efficiency Analysis

5.1 Why a separate cruise analysis was used

Cruise fuel-use patterns are easier to interpret when highly dynamic climb and descent intervals are excluded. The analysis therefore focused on observations classified as cruise and restricted attention to the main observed cruise altitude region.

Importantly, the optimization search was limited to cells with adequate empirical support rather than allowing the fitted model to extrapolate far outside the observed data.

This distinction is important because a machine-learning model can produce apparently favorable predictions in regions where little or no training data exist. Such extrapolation can be misleading.

5.2 Training-data supported region

Within the main cruise region and after support filtering, the most favorable model-supported area was around **36,000 ft and groundspeed 390** in the training analysis.

The lowest observed supported training cell was around **35,500 ft and groundspeed 400**, with an observed median fuel rate near **32.3 kg/min**.

Because the model and the observed cell statistics do not identify exactly the same point, the results are better interpreted as a neighborhood of low fuel-rate observations rather than a unique optimum.

5.3 Out-of-sample validation of the cruise analysis

To reduce selection bias, the cruise search was repeated on the untouched validation flights.

The validation subset contained:

Table 8: Out-of-sample validation statistics for the cruise efficiency analysis.

Metric	Value
Validation cruise observations	345.000
Validation cruise flights	47.000
Supported validation cells	15.000
Cell-level RMSE	2.458
Cell-level MAE	1.974
Spearman rank correlation	0.693
Mean absolute rank difference	2.670

The validation data contained 345 cruise observations from 47 flights. After support filtering, 15 validation cells remained.

The cell-level out-of-sample RMSE between model predictions and observed fuel-rate summaries was approximately **2.46 kg/min**, with MAE approximately **1.97 kg/min**. The Spearman rank correlation was approximately **0.693**, indicating that the relative ordering of supported cells was reasonably preserved out of sample.

The lowest observed validation cell occurred around:

- altitude = **36,000 ft**;
- groundspeed = **410**;
- observed median fuel rate = **37.4 kg/min**.

The lowest model-predicted validation cell occurred around:

- altitude = **36,000 ft**;
- groundspeed = **420**;
- observed median fuel rate = **39.8 kg/min**.

Table 9: Selected supported cruise cells from training and validation analyses.

Analysis	Altitude (ft)	Groundspeed	Fuel rate (kg/min)	Flights
Training model-supported region	36000	390	NA	NA
Training observed supported region	35500	400	32.3	NA
Validation observed lowest cell	36000	410	37.4	4

Analysis	Altitude (ft)	Groundspeed	Fuel rate (kg/min)	Flights
Validation model lowest cell	36000	420	39.8	4

5.3.1 Optimization heatmaps

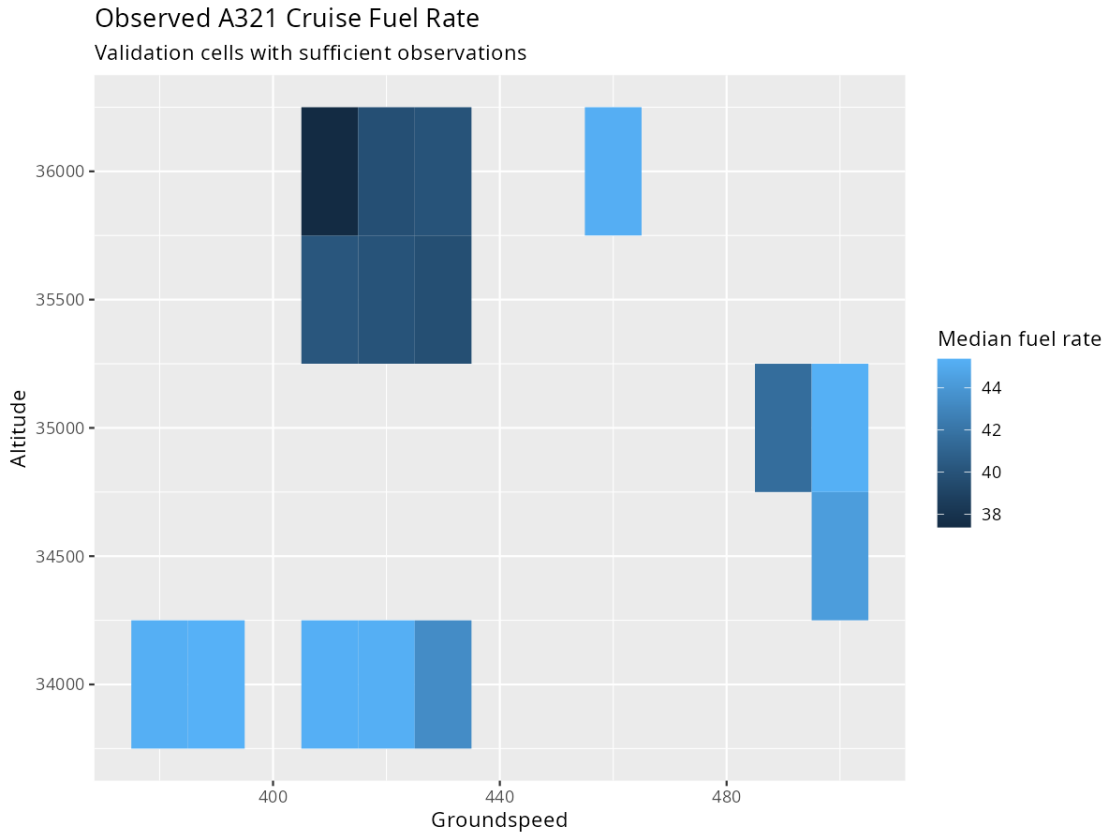


Figure 7: Observed supported cruise fuel-rate surface in the training analysis.

These heatmaps should be interpreted as descriptive model-assisted analyses of the available data. The supported low-fuel region is approximately clustered around the mid-35,000 to 36,000 ft altitude range and groundspeeds in the low 400s, but the result should not be interpreted as a prescriptive aircraft operating recommendation.

6 Discussion

6.1 Main findings

The analysis produced four main findings.

First, fuel rate differed substantially by flight phase. Climb intervals had the highest mean fuel rate, cruise intervals were intermediate, and descent intervals had the lowest mean rate.

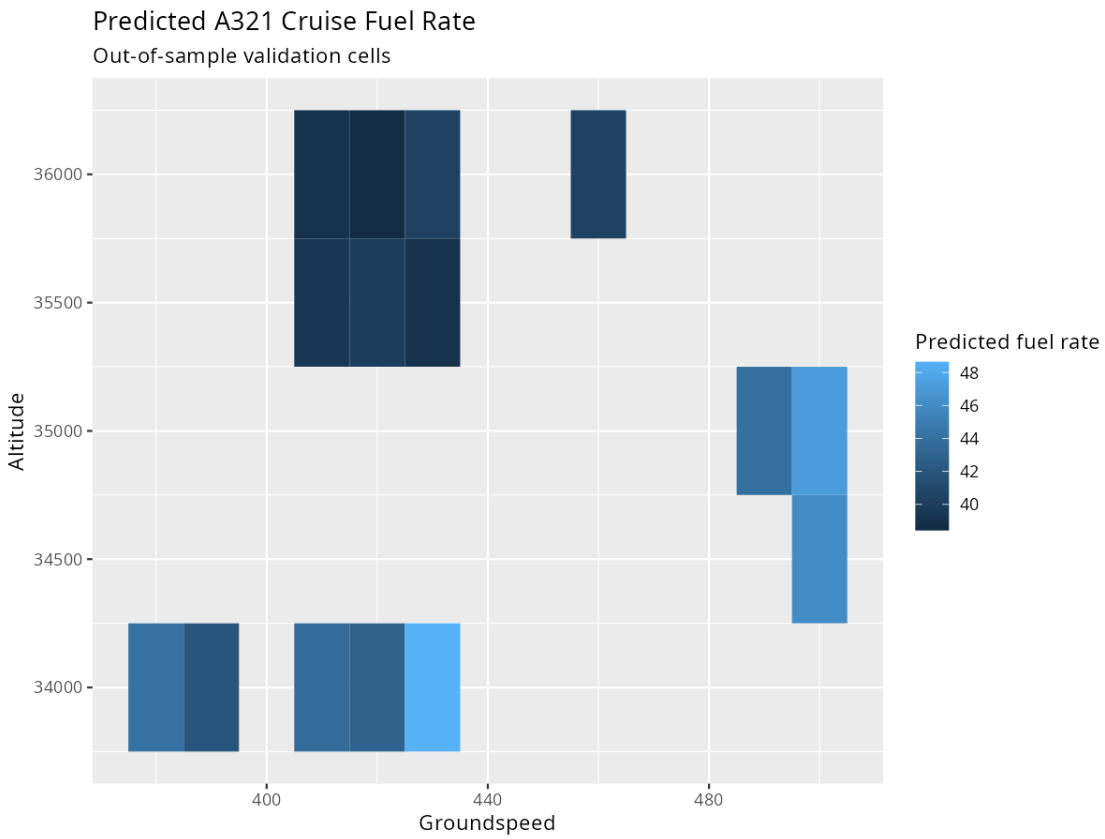


Figure 8: Random-forest predicted fuel-rate surface in the supported cruise region.

Second, nonlinear models improved validation performance relative to simple baselines and linear regression. The random forest produced the lowest validation RMSE among the tested models.

Third, the predictive contribution of altitude change and vertical-rate-related variables was substantial in the random forest. This is consistent with the descriptive observation that dynamic changes in the aircraft's vertical state are associated with fuel-rate variation.

Fourth, the cruise efficiency analysis produced a relatively stable low-fuel region in the held-out validation data. The fact that the supported validation ranking had a Spearman correlation of about 0.693 suggests that the model retained useful information about relative fuel-rate conditions beyond the training flights.

6.2 Interpretation of the model

The random forest should primarily be interpreted as a predictive model. Its feature-importance values indicate which variables contributed strongly to prediction within the fitted sample, but they do not establish that changing one variable while holding all others fixed would necessarily cause a corresponding change in actual fuel consumption.

Operational aircraft fuel burn depends on many factors that are not included here, including aircraft mass, engine condition, temperature, wind, air traffic constraints, route structure, and airline operating procedures. The current dataset and feature set therefore support data analysis, but not a complete physical or operational fuel-burn model.

7 Limitations

This project has several limitations.

7.1 Limited A321 sample

Although the original dataset contains many flights and aircraft types, the A321 subset used here contains only 285 matched flights, with 283 flights represented in the final modeling data. This is substantially smaller than the full dataset.

7.2 Interval-level target

The target is an interval-level fuel rate rather than total fuel burn for an entire flight. The interval duration and telemetry quality therefore affect the measurement directly.

7.3 Short intervals

A large number of observations are shorter than five minutes, and 165 observations are shorter than one minute. Short intervals can generate unstable fuel-rate estimates when a relatively large fuel value is divided by a very small duration.

7.4 Missing trajectory variables

Several raw trajectory variables were highly incomplete. Raw CAS, TAS, and Mach were therefore not used directly in the final models. Derived speed and vertical-rate measures were used instead.

7.5 Limited operational context

The dataset does not provide all of the variables needed for an operationally complete fuel-consumption model. The analysis should therefore be viewed as an empirical machine-learning study rather than a replacement for certified aircraft performance calculations.

7.6 Optimization should not be treated as a flight-planning rule

The altitude/speed analysis identifies low fuel-rate regions in this dataset. It does not account for safety requirements, air-traffic restrictions, weather, payload, reserves, airline procedures, or aircraft-specific operational constraints.

8 Conclusion

This project developed a machine-learning workflow to predict interval-level fuel consumption rates for Airbus A321 flights using trajectory and fuel telemetry data.

After data matching and quality filtering, the final analysis included 2,273 interval observations from 283 A321 flights. Exploratory analysis showed clear differences in fuel-rate behavior across flight phases, with higher fuel rates during climb and lower rates during descent. Short-duration intervals also showed greater variability in fuel-rate estimates.

Three predictive models were evaluated: linear regression, random forest, and gradient boosting. The random forest achieved the strongest validation performance, with an RMSE of 19.77 kg/min, an MAE of 10.05 kg/min, and an R-squared value of 0.368. Its performance remained strongest in the robustness analysis restricted to intervals of at least one minute.

A separate cruise analysis examined altitude and groundspeed combinations within regions supported by sufficient observed data. The held-out validation analysis showed that lower observed fuel-rate cells were concentrated around approximately 35,500–36,000 ft and groundspeeds in the low 400s. These results indicate patterns present in the dataset rather than a unique operational optimum.

Overall, the study demonstrates that machine-learning models can capture useful relationships between A321 flight characteristics and observed fuel consumption. However, the current analysis does not include important operational factors such as weather, aircraft mass, traffic constraints, or airline procedures. Therefore, the optimization results should be interpreted as observational and model-based findings, not as operational flight recommendations.

Future work could expand the A321 sample, incorporate weather and aircraft-mass information, predict total flight fuel rather than interval fuel rate, develop richer temporal features, and evaluate additional models using stronger cross-validation strategies.

9 References

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